

Analytic Verification AV-11 - Rev B

1. Objective

Verify the released nominal magnetic circuit against closed-form analytic calculations under the linear quasi-static assumptions.

2. Test configuration

Parameter	Value
a	0.025 m
l	0.150 m
delta	0.00150 m nominal
mu_r	1200
sigma	0.08
N_exc / N_meas	600 / 50 turns
Frequency	50 Hz
Current ramp	0 A before 0.10 s; linear to 2.0 A RMS at 0.50 s; hold through 1.0 s

3. Analytic checks at 2 A RMS

Quantity	Expected
Total core reluctance	5.289249e+05 A/Wb
Air-gap reluctance	1.909859e+06 A/Wb
Equivalent gap/leak reluctance	1.757071e+06 A/Wb
Total core flux	5.249354e-04 Wb
Useful air-gap flux	4.829406e-04 Wb
Leakage flux	4.199483e-05 Wb
Core B	0.839897 T
Gap B	0.772705 T
Core magnetic drop	277.651 A-turn
Gap magnetic drop	922.349 A-turn
Exciting induced U	98.948 V RMS
Measuring induced U	7.586 V RMS

4. Acceptance

MMF balance residual shall be within 0.5% of $N \cdot I$. Useful/core flux ratio shall be 0.92 within 0.002 absolute. Flux, B, H and induced-voltage magnitudes shall each agree within 1% of the released analytic result.